Economic importance of Frog

- Frog is an important animal in the **food chain**; it helps to maintain our ecosystem. So '**frogs should be protected'.**
- Frog are beneficial to man, since they feed on insects and helps in reducing insect pest population.
- Frogs are used in traditional medicine for controlling **blood pressure** and for its **anti aging** properties.
- In USA, Japan, China and North East of India, frogs are **consumed** as delicious food as they have high nutritive value.

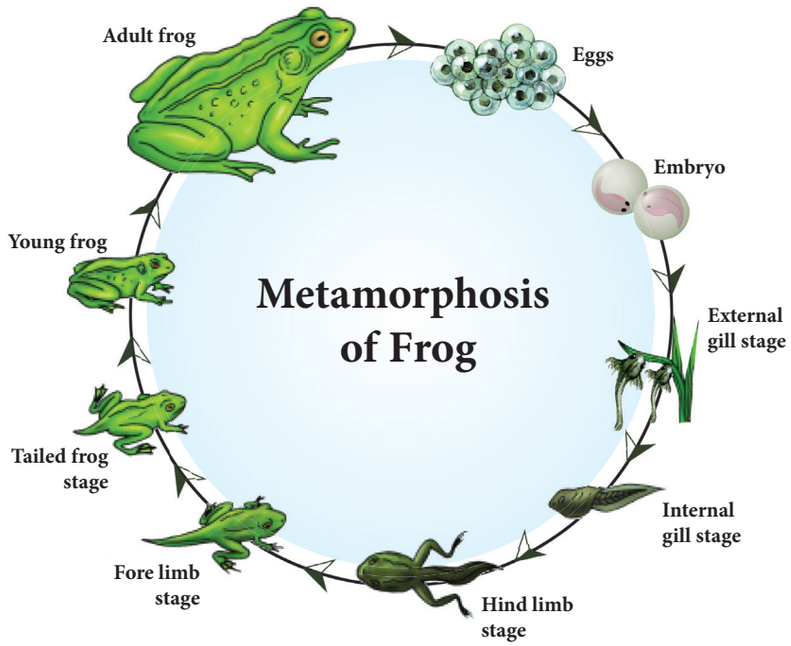


Figure 4.25 *Rana hexadactyla* - Metamorphosis

Summary

Earthworm, Cockroach and Frog show characteristic features in body organization. *Lampito mauritii* (earthworm) is commonly found in Tamil Nadu, its body is covered by cuticle. It has a long and cylindrical narrow body which is bilaterally symmetrical. All segments of its body are alike except the 14 to 17 segments, which are thick and dark and glandular, forming the clitellum. This helps in the formation of cocoons. A ring of S-shaped chitinous setae is found in each segment. These setae help in locomotion. Earthworm's development is direct and no larva is formed during development.

Cockroach is a typical cosmopolitan insect and exhibits all the fundamental characteristics of class Insecta. The body of the cockroach is compressed dorso-ventrally, bilaterally symmetrical, segmented and divisible into three distinct regions – head, thorax and Abdomen. The photoreceptor organ of the cockroach consists of a pair of compound eyes with mosaic vision. Segments bear jointed appendages. There are three throacic segments of each bearing a pair of walking legs. Two pairs of wings are present, one pair each on 2nd and 3rd segment. There are ten segments in abdomen. Fertilization is internal. The development of cockroach is gradual through nymphal stages (paurometabolus).

Frogs are cold blooded vertebrates – Poikilotherms. Skin is smooth and moist, Red blood corpuscles are nucleated. Eggs are laid in water. The larvae pass through an aquatic stage before metamorphosing into adult.

TN_GOVT_XI_Zoology_Ch04.indd 69 09-01-2020 5.14.31 PM